

Dermatology and restorative medicine

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The skin is the largest organ of the human body, with a surface area of approximately 2m^2 .¹ It is a barrier between the internal and external environment and has multiple specialized functions including thermoregulation, hydration, immunological surveillance, and protection against external chemical, mechanical, and pathogenic insults.

Skin aging, like aging of the other organs, is characterized by a progressive loss of functionality and regenerative potential. It is a complex and heterogeneous process affecting the different constituents of the skin that is influenced by both intrinsic and extrinsic factors. The aging process is proposed to consist of nine concomitant hallmarks including genomic instability, epigenetic alterations, telomere shortening, loss of proteostasis, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, and altered intercellular communication.²

The skin's capacity to efficiently mirror age-associated changes in the entire human body further highlights the importance of elucidating the mechanisms of skin aging.³ Thus, skin aging is not only a physiologic phenomenon but also a health risk, resulting in increased skin fragility, delayed and impaired wound healing, and increased incidence of infection and skin

cancers. As the proportion of the aging population in industrialized countries continues to increase, the dermatological focus gradually shifts towards improving the cosmetic outcomes of skin aging and reducing the morbidity associated with aging skin. In this chapter, we focus on the pathomechanisms and clinical features of skin aging together with prevention and treatment recommendations.

Intrinsic and extrinsic skin aging

The structural and functional changes that occur in the skin with age can be attributed to the synergistic and deleterious consequences of intrinsic and extrinsic factors. Although it is difficult to distinguish the respective contributions of intrinsic and extrinsic factors to the aging phenotype in most sites, skin aging in photoprotected sites is mainly attributed to intrinsic factors, whereas intrinsic and extrinsic factors are superimposed in sun-exposed skin.

Intrinsic aging

Intrinsic or chronological aging is an inevitable, genetically determined process that occurs in all organs with age due to the accumulation

of the oxidative damage of reactive oxygen species (ROS), mitochondrial DNA mutations, progressive telomere shortening, and cellular senescence.⁴ It is associated with genetic factors, ethnicity, metabolism, and hormonal changes. Collagen synthesis declines by 30% in the first 4 years of menopause, then by 2% annually. In particular, the deficiency in estrogens and androgens causes dryness, wrinkling, epidermal atrophy, collagen breakdown, and loss of elasticity, which can be at least partially repaired by systemic hormone replacement therapy (HRT) or local estrogen treatment.

Intrinsically aged skin appears macroscopically thin and atrophic and exhibits fine wrinkles as well as loss of underlying fat, reduced elasticity, and prominent xerosis, often accompanied by pruritus. Development of benign vascular formations such as cherry angiomas and benign skin tumors such as seborrheic keratoses, hair graying, and hair loss are among other features of intrinsic skin aging.

Extrinsic aging

In addition to the inevitable chronological decline, the skin also undergoes extrinsic aging due to its interface with the environment. Extrinsic aging depends on the intensity and duration of exposure to environmental factors and on the skin type. Air pollution and cigarette smoke are well-known external factors that accelerate skin aging; however, the most significant extrinsic aging factor is still ultraviolet radiation (UVR).⁵

Photoaging is the superposition of chronic UVR-induced damage on intrinsic aging. UVR exerts its manifestations through induction of DNA damage and alterations through UV-mediated ROS, including interference with the cutaneous immune system.⁶ UVB is the major cause for direct DNA damage and induces inflammation and immunosuppression, while UVA may have a greater role in inducing the dermal changes of photoaging due to its ability

to penetrate deeper into the dermis. In addition to differences in their depths of penetration, UVA and UVB also produce different cellular effects that are likely influencing their photoaging effects. UVR affects all components of the skin, including the keratinocytes, melanocytes, the vasculature, the dermal connective tissue, and possibly even the subcutaneous fat. Chronic and repetitive inflammatory responses to UV light results in upregulation of extracellular matrix (ECM)-degrading proteases and downregulation of collagen synthesis, with subsequent loss of collagen and deposition of abnormal degenerative elastotic material within the dermis (actinic/solar elastosis) and advanced glycation elements (AGE) accumulation over time. Histopathologically, photodamaged skin displays variable epidermal thickness, dermal elastosis, decreased and fragmented collagen, increased matrix-degrading metalloproteinases, inflammatory infiltrates, and vessel ectasia.

Clinical signs of cutaneous photoaging include rhytids, lentigines, keratoses, telangiectasias, loss of translucency, loss of elasticity, and a sallow color. Skin type plays an important role in determining the visible effect from photoaging. Photoaging is much less pronounced and has a delayed onset in individuals with more darkly pigmented skin due to the protective effect of eumelanin against the cumulative effects of photoaging. Individuals with Fitzpatrick skin types I and II show atrophic signs with fewer wrinkles, focal depigmentation (guttate hypomelanosis), and dysplastic changes, such as actinic keratoses. In comparison, type III and IV skin display hypertrophic features with thick, course wrinkles, leathery skin, lentigines, and an overall “bronzed” appearance. A yellow cobblestone appearance of the skin may also be present resulting from the elastosis in photoaging.⁷ Additionally, changes in the dermal vasculature following UV radiation may also lead to dilatations, which clinically appear as telangiectasias and easy bruising after simple trauma.

Basal cell (BCC) and squamous cell carcinomas (SCC) occur almost exclusively on sun-exposed skin of fair-skinned individuals.

Chronic exposure to UVR also increases the risk of both melanoma and nonmelanoma skin cancers.⁸ For BCCs, intermittent intense episodes of UV exposure and sunburns at any age appear to increase risk, whereas cumulative long-term UV exposure and childhood sunburns increase the risk for developing SCCs and actinic keratosis (AK). Sun exposure early in life appears to have a greater influence on subsequent skin cancer risk than exposure to sun at a later age.

Further relevant exogenous factors for skin aging are exposure to tobacco smoke, environmental pollutants, infrared radiation, temperature-related factors, and miscellaneous lifestyle components such as stress, nutritional factors, and even lack of sleep and overall health.⁹ Recently, the term skin exposome has been used to describe the total of potentially deleterious, age-inducing external factors to which skin is exposed during a lifetime.⁹ Such factors, singly or in combination, may have the potential to conspire in a manner that affects the epigenome governing the transcriptional integrity of skin cell DNA, thus pathologically modifying cells, including the stem cells. Exposure to tobacco smoke accelerates extrinsic aging processes and targets mainly the elastin network of the skin.^{10,11} It also induces MMP-1-related collagen degradation.¹² Cigarette smokers' wrinkle formation depicts a distinctive pattern with prominent perioral lines and sharply contoured crow's feet, known as "smoker's face."

Histopathological and physiological characteristics of aging skin

The skin consists of several compartments with different functions. Characteristic structural changes observed in the epidermis, dermis,

and hypodermis with age contribute to the visual signs of skin aging.

Epidermis

The epidermis plays a major role in the maintenance of homeostasis and protects the organism from physical, chemical, and infectious insults. The epidermis consists of different cell types including immunoregulatory Langerhans cells, photoprotective melanocytes, and keratinocytes, which predominantly constitute 95% of the epidermis. Changes in the epidermis in aged skin are characterized by reduced mitotic activity of keratinocytes in the basal layer associated with epidermal atrophy and decreased epidermal turnover rate with an altered permeability of the stratum corneum. Epidermal thickness decreases approximately 6.4% per decade.¹³ The thickness of the epidermis of photoaged skin is greater than that of chronological aged skin, whereas epidermal atrophy is observed in severely photoaged skin. Slowing of the epidermal turnover rate and cell cycle lengthening coincides with a slower wound healing and less effective desquamation in older adults.

The stratum corneum, the outermost layer of the epidermis, consists of corneocytes arranged in compact sheets that are embedded in an ECM derived from lamellar granules forming lipid-rich membranes. Although the thickness of the stratum corneum does not change with age, stiffening of this layer with increasing cellular cohesion and decreased concentration of intercellular lipids occurs with aging.¹⁴ Transepidermal water loss through the stratum corneum is also decreased.¹⁴ The decrease in transepidermal water loss with aging might be a result of the changes in the stratum corneum. These changes, in addition to alterations in sebum secretion, likely play a role in the xerotic phenotype of aged skin. Skin hydration decreases with age due to the reduction of

hygroscopic natural moisturizing factors, diminished lipid-processing enzymes, and an altered lipid composition in the stratum corneum.¹³

One of the first alterations that occurs with age and that appears independent of repeated exposure to UV light is thinning of the epidermal layer associated with loss of epidermal rete ridges and their reciprocal inter-digitation with capillary-rich dermal papillae, which lead to reduced nutrient support to the epidermis. The attenuation of the rete ridges also results in a decreased capacity of the skin to resist shearing forces, commonly illustrated by the comparative difficulty in removing a band aid from the skin of an elderly person versus the skin of a child without incurring epidermal disruption or associated hemorrhage.

In addition to age-related changes in keratinocytes, the number of melanocytes decreases from 8% to 20% per decade, while the number of senescent melanocytes increases with age and contributes to the pallor of aging skin and loss of hair pigmentation.¹⁵ Although the number of senescent melanocytes increases with age, surprisingly the number of senile lentigos also increases. In photoaged skin melanogenesis is also upregulated for neutralization of ROS induced by UVR exposure.¹⁶ The number of Langerhans cells in the epidermis markedly decreases in both intrinsically and extrinsically aging skin, which leads to impaired skin immunity.¹⁷

Dermis

The human dermis has a dynamic architecture, containing a multitude of cell types such as fibroblasts and immune cells, and a composite network of ECM proteins, which provide skin hydration, turgidity, and viscoelastic properties to withstand external compression forces, mechanical strength and elastic recoil. In adult skin, the ECM is comprised of collagen network, the elastic fiber network, proteoglycans, glycoproteins, glycosaminoglycans (GAGs), and

water. The interactions of water with GAGs such as hyaluronic acid (HA) generate a large osmotic swelling pressure conferring skin stiffness and resistance to deformation.

From the age of 50 years, the quality of the dermis gradually deteriorates. Dermal thickness decreases with age in parallel with progressive attenuation and loss of dermal-epidermal junction undulations, which are correlated with reduced skin elasticity, extensibility, and increased laxity.¹⁸ Concurrently with dermis atrophy, the thickness of collagen bundles decreases and the space between bundles increases, which leads to a reduction of tissue density.¹⁹

Dermal fibroblasts

Dermal fibroblasts are responsible for the synthesis and organization of the ECM in the skin. The dermis of chronologically aged skin has fewer fibroblasts.²⁰ There are also alterations in fibroblast functions including contractile capacity, secretory profile with an upregulation of pro-inflammatory cytokines, pro-oxidant factors, and degradative matrix metalloproteinases (MMPs).²¹ Fibroblast senescence contributes to skin aging by secreting a senescence-associated secretory phenotype (SASP), which decreases proliferation by impairing the release of essential growth factors (GFs) and enhances degradation of the ECM by activating MMPs.²¹ The increase in MMP secretion contributes to the atrophy of the ECM networks.

Collagen network

Collagen fibers are the major constituents of dermal ECM that correspond to 70%–80% of the skin's dry weight. The adult dermis contains the majority of fibrillary collagens, including type I (80% of all collagens), type III (15%), and type V collagens (5%). Types I and III collagens constitute the principal skeleton that drives dermal matrix organization. Alterations in collagen play an integral role in the skin aging process. In the dermis of chronologically aged skin, density, thickness, and the degree of

organization of dermal collagen, which are the main contributors to overall tissue stiffness and resilience, are all decreased.²² Collagen-degrading enzymes (e.g., MMPs) are upregulated during both photoaging and intrinsic aging, mainly via the production of ROS. Collagen synthesis decreases resulting in a shift in the balance between synthesis and degradation.²³ Some of these alterations may be explained by an accumulation of advanced glycation end products (AGEs), especially carboxy methyl lysine, which is proposed to induce conformational modifications of collagens.²³

Elastic fibers

Elastic fibers, which are composed of filaments called microfibrils and an amorphous component, show structural changes during the aging process. UVR triggers remodeling/degradation of elastic fibers mostly regulated by MMPs and abnormal localization of altered elastin fibers in the upper dermis of photo-damaged skin, commonly referred to “solar elastosis.” Solar elastosis is the most striking histological feature of photo-damaged skin and represents a tangled mass of degraded elastic fibers, disorganized tropoelastin and fibrillin located in the upper dermis. Although the mechanism is not understood fully, it could be the result of both degradation and de novo synthesis of elastic fibers due to UV exposure. This abnormal elastotic material is nonfunctional and confers neither elasticity nor resilience to the skin. Thus, recovery from mechanical depression takes only minutes in young skin, whereas it can extend more than 24h in the elderly.

Glycosaminoglycans (GAGs)

GAGs and particularly HA are major components of the cutaneous ECM involved in tissue repair. The level of GAGs increases in photoaged skin, but while young skin is well hydrated because most of the water is bound to proteins, there is a shift towards “tetrahedron water” in aged skin. This has poor hydration and turgor

capacity and contributes to the xerotic appearance of photoaged skin. Extrinsic skin aging is characterized by distinct homeostasis of HA. The elucidation of the role of HA homeostasis in extrinsic skin aging may offer an additional approach in handling cutaneous aging.²⁴

Vasculature

Skin aging is also associated with a decrease in cutaneous blood flow, especially in sites exposed to the environment. Both intrinsically aged and photoaged skin show an age-dependent reduction of cutaneous vessel size, whereas only photoaged skin exhibits significantly reduced numbers of dermal vessels with a particular loss of vertical capillary loops in the superficial part of the dermis.²⁵ In addition, histopathologically enlarged vessels due to thinning of the vessel walls in severely photodamaged skin clinically manifest as telangiectasias and senile purpura. The reduction in skin perfusion results in skin pallor, depleted nutrient exchange, impaired thermoregulation, and impaired dermal immune cell homeostasis.²⁶

Hypodermis

The hypodermis consists mainly of fat and acts as an insulator and shock absorber. With age, there is a loss of subcutaneous fat, and the skin becomes thinner and less resistant to trauma. The distribution of subcutaneous fat also changes with a decrease in the face and hands and an increase in the thighs and abdomen.

Skin appendages

The appendages of the skin include sebaceous, apocrine, and eccrine glands in addition to hair follicles. These appendages are responsible for several crucial roles of the skin, including re-epithelialization after injury, thermoregulation, and endocrine function. The sebaceous

glands not only produce sebum but also have antibacterial activities, inflammatory functions, and roles in hormone metabolism. Although the number of the sebaceous glands is not affected by age, the rate of sebum production decreases and sebaceous glands become dysfunctional and undergo hyperplasia in aged skin as a compensatory response to falling androgen levels. Clinically, in addition to sebaceous gland hyperplasia, which may be cosmetically bothersome to the patient, these alterations contribute to xerosis, which is one of the most common manifestations of the aged skin.

Mechanisms of skin aging

Genetic factors

Genetic differences may play an important role in skin aging. Genome-wide association studies have indicated that single nucleotide polymorphisms (SNPs) associated with the MC1R gene may play a role in skin aging regardless of melanin-related photoprotection.²⁷ Epigenomic alterations that reorganize chromatin and alter gene expression are also known to contribute to cellular aging in organisms.²⁸ Overall, aging tissues are replete with epigenetic anomalies, including reduction in global heterochromatin, nucleosome loss and remodeling, alterations in histone marks, relocalization of chromatin-modifying factors, and global hypomethylation of DNA with hypermethylation of CpG islands. However, it remains unclear whether such changes are the consequence of other processes that drive aging or if they are directly upstream to gene expression pathways that regulate the acquisition of the gradually evolving disease known as aging.

Microbiota

The skin plays host to a diverse ecosystem, housing up to a million bacteria per square centimeter of surface area, in addition to viruses

and fungi. Clearly, the interplay between the microbiome, skin, and the immune system represents a delicate balance that may be more susceptible to disruption with age. Interestingly, there are significant differences in microbiome populations among young and old skin. Metagenomic studies comparing microbiomes from young and old skin found that the overall microbiota structure was different between the two groups and skin bacterial diversity increased with aging.¹⁶

A potential link between the microbiome and the development of cutaneous malignancies has also been proposed. Although there is no definitive evidence currently linking the skin microbiome to skin cancers, this is an exciting emerging area of research.

Stem cells

Stem cells communicate with organ systems to adjust immune response, provide signals for differentiation, migration, and enzymatic reactions, and to equilibrate the regenerative demands of balanced tissue homeostasis. Stem cell numbers do not necessarily decline with age; however, their functional role to produce differentiated progeny has been documented to be disturbed. There is the possibility of impairment of stem cell mobilization with age, as well as the inability to respond to proliferating signals. Multiple mechanisms are involved in stem cell exhaustion including expressing lower levels of the stem cell markers β 1-integrin and melanoma chondroitin sulfate proteoglycan, which are correlated with a greater self-renewal capacity.

Age-related skin stem cell depletion impacts negatively on both skin structure and functions and leads to changes such as epidermal atrophy and delayed wound healing. It is likely that the microenvironment for which stem cells are responsible no longer provides a safe harbor to other cells critical to the protective function of the integument.

Epidermal stem cells reside within the basal cell layer and give rise to transient amplifying cells and differentiating progenitors. Epidermal stem cells are considered unique in comparison with other adult stem cells in their ability to resist aging. They show no effects associated with increased ROS levels, perhaps as a result of maintaining high levels of superoxide dismutase. Despite these factors, the epidermal turnover rate is 28 days in young individuals, whereas it varies between 40 and 60 days in the elderly. Dermal stem cells are also of great importance for skin homeostasis because they produce the progeny responsible for extracellular matrix synthesis and GFs.

Clinical features of skin aging

Aging, both intrinsic and extrinsic, induces several skin manifestations. There are a number of important skin functions that deteriorate with increasing age, including epidermal regeneration capacity, synthesis of sebum and sweat, dermoepidermal adhesion, wound healing, thermoregulation, and the speed of natural elimination of potentially hazardous chemical factors. We summarize common age-associated skin changes in the sections that follow.

Changes in skin appearance

Important aspects of skin aging that can be addressed occur in skin hydration, barrier function, content of the matrix, pigmentation, and antioxidant capacity. Skin aging is usually assessed by examination of individual skin features including wrinkles, pigment irregularities, sagging, or global evaluation of appearance. The underlying causes of wrinkles are multiple and include UVR-induced damage, loss of skin elasticity, repetitive movement of underlying facial musculature, changes in volume from fat redistribution, and bone as well as cartilage resorption. Skin laxity occurs as a result of a decreased underlying framework, a thinned

and damaged dermis, and a breakdown in elastic and collagen fibers. As the facial soft tissue ages, both the skin and the underlying supporting structures age as well and lose their inherent resiliency and ability to resist stretching. Under the effects of gravity these changes impair skin laxity.

Impaired wound healing

Skin aging is associated with an impaired wound healing process, which can lead to leg or decubitus ulcers. The fundamental underlying defect in all age-related healing deficiencies may relate to functional and numerical depletion in skin stem cells. Age-associated deficits in microvascular anatomy and function contribute to impaired wound healing in the aged population.

Age-related immune system alterations

The whole skin, excluding the immune-privileged transient portion of anagen hair follicles, is a potent immune-competent tissue. The density and ability to migrate from the epidermis in response to cytokines of antigen-presenting Langerhans cells in the skin decreases greatly in the elderly even in sun-protected sites. Similarly, T lymphocytes become less responsive to specific antigens with aging.²⁹ Aging skin also appears to have a reduced ability to produce certain cytokines (e.g., interleukin-2), while the production of others (e.g., interleukin-4) is increased. The consequence of these changes is a reduced intensity to delayed hypersensitivity reactions, increased susceptibility to photocarcinogenesis, autoimmune diseases (e.g., bullous pemphigoid), and opportunistic skin infections.

Skin cancer

UVR-induced cellular DNA alterations play a major role in cutaneous carcinogenesis. Additionally, immune senescence may also result in

an increase of infection-associated cutaneous malignancies with age, including HPV-related SCC and Merkel cell polyoma virus-associated Merkel cell carcinoma. UVR acts synergistically with oncogenic viruses by inducing driver mutations in viral oncogenes and promoting immune tolerance in the skin. This interaction may provide possible targets for prevention and treatment of these cancers.³⁰

Prevention and treatment methods

Skin health is considered one of the principal factors representing overall “wellbeing” and the perception of “good health” in humans. Awareness of the pathophysiology of age-associated skin alterations and diseases as well as of preventive measurements to avoid skin damage is the first step for successful, healthy aging.

Aging of the entire face is associated with gravity impact, muscle action, loss of volume, diminishing and redistribution of superficial and deep fat, and loss of bony skeleton support. Taken together, these lead to facial sagging and changes in the face’s shape and contour. The desired therapeutic antiaging effect of the skin is a continuous, step-by-step process that combines various methods of skin biorevitalization and rejuvenation, augmentation, and restoration of each skin layer individually.

The aim of any skin antiaging therapy is to achieve a healthy, smooth, blemish-free, translucent, and resilient skin. Several strategies have been developed during the last decade that can promote skin youthfulness and may delay skin aging. These strategies can be grouped under three categories: preventive measurements, therapeutic agents, and minimally Invasive procedures. Avoiding the exogenous factors of aging, sun protection, correction of lifestyle and habits, and daily skin care routine are the mainstay of preventive measurements. Topical and systemic therapeutic agents with antiaging properties, injectables, and energy-based procedures will be discussed below.

Combination therapy that addresses dynamic lines, volume loss, and photoaged skin will give optimal results. The skin antiaging strategies for reversing the dermal and epidermal signs of photo and chronological aging can be grouped as shown in [Table 1](#).

Daily skin care

A healthy and functioning skin barrier is an important protector against dehydration and penetration of various microorganisms, allergens, irritants, ROS, and radiation. Daily skin care may increase skin regeneration, elasticity, and smoothness and thus temporarily change the skin condition. Some products may promote the natural synthesis of primary structural constituents of the skin such as collagen and elastin.

Sun protection

UV exposure is the main driving factor of extrinsic aging and causes the appearance of wrinkles and uneven pigmentation. It is crucial to protect the skin from the damaging effects of UVR and to counteract the deleterious effects of oxidative stress. Sun protection strategies include avoiding sun exposure during the peak sun hours of 10 a.m. to 4 p.m., and wearing clothing, hats, and sunglasses that block UVR. Daily sunscreen should be used in addition to these sun protection strategies. Sunscreens that are broad spectrum (protective against both UVA and UVB radiation) and have a sun protection factor (SPF) of 30+ are recommended.³¹ The total amount of sunscreen applied to the skin should be a quantity of 2 mg/cm². Additionally, sunscreen should be applied 15–30 min before sun exposure and reapplied every 2–3 h especially after swimming and/or sweating. Obviously, indoor tanning beds should be avoided.

Cessation of smoking

The aging effects of smoking can appear in as little as 10 years after one begins smoking. Nicotine narrows blood vessels, causing less oxygen

TABLE 1 Skin antiaging approaches.

Preventive medicine (avoiding exogenous factors of aging, correction of lifestyle and habits)	Smoking Pollution Solar UV irradiation Stress Nutrition, diet restriction, and alimentary supplementation Physical activity Control of general health
Cosmetologic care	Daily skin care Sun protection Esthetic noninvasive procedures
Topical medical agents	Antioxidants (vitamins, polyphenols, and flavonoids) Cell regulators (retinols, peptides, and growth factors (GFs))
Minimally invasive procedures	Chemical peelings Visible light devices Intense pulsed light (IPL) Ablative and nonablative laser photo- rejuvenation Radiofrequency (RF) Injectable skin biostimulation and rejuvenation Prevention of dynamic wrinkles Correction of static, anatomic wrinkles Restoration (redistribution) of fat and volume loss, skin augmentation, and contouring
Systemic agents	Hormone replacement therapy Antioxidants

and nutrients to be delivered to the skin. The postischemic reperfusion injury, as well as other smoke compounds, generates damaging free radicals.

Adoption of a healthy lifestyle

A healthy lifestyle is the first line of prevention to avoid the accelerated appearance of signs of aging. Eating a healthy, well-balanced diet along with taking antioxidant supplements, such as vitamins C and E, may help prevent skin damage that leads to premature aging. These vitamins have antioxidant properties that neutralize damage by free radicals. Recent evidence suggests that the Mediterranean diet may be helpful to prevent early aging. Foods with high

glycemic index cause production of AGEs, which damage collagen and lead to saggy skin. Alcohol consumption can dehydrate the skin and may, over time, cause skin damage. Reducing alcohol consumption is also beneficial. Exercise can improve circulation, boost the immune system, and improve mitochondrial functionality. Proper sleep helps to refresh the body and may give the skin a more youthful look.

Topical agents

Topical retinoids

Tretinoin and tazarotene are the two topical retinoids (vitamin A derivatives) that are currently FDA approved for the treatment of

photoaging. On a molecular level, retinoids have been postulated to exert their effect via their nuclear retinoic acid receptors, which act as transcription factors in gene regulation. In addition, retinoids have been shown to increase collagen production, upregulate fibroblast proliferation, and block dermal matrix degradation following sun exposure.^{32,33}

Cosmeceuticals

Cosmeceuticals encompass a heterogeneous category of nonprescription topical products including antioxidants, cell regulators, and hydroxy acids.

Antioxidants are proposed to reduce collagen degradation by reducing the concentration of free radicals in the tissues. Many of the antioxidant substances used in antiaging cosmetics are chemical molecules (vitamin C, vitamin E, glutathione, coenzyme Q10, alpha lipoic acid) or molecules of plant origin such as carotenoids, flavonoids (flavones, flavanones, anthocyanins, catechins), stilbenoids (resveratrol, imbricatin), and polyphenols (ellagic acid, gallic acid, chlorogenic acid, tannins). Vitamin C is a popular antioxidant that has been shown to upregulate collagen synthesis in skin as well as prevent erythema after UV exposure. Numerous studies using doses from 5% to 17% for 12 weeks to 6 months' duration report a clinically significant reduction in photoaging along with improvements in facial texture and wrinkle appearance. Unfortunately, the main drawbacks to vitamin C use include its poor penetration after topical use and its inherent instability resulting in a short half-life. Thus, vitamin C products are usually supplemented with other antioxidants including vitamin E to compensate for these shortcomings. Vitamin E is produced by sebaceous glands via its alpha-tocopherol and gamma-tocopherol forms. It is found in many skin care products because of its photoprotective effects in UV and infrared settings, excellent moisturizing capabilities, and functions as a preservative in cosmetics. On top of its antioxidant abilities, it may also possess

antiinflammatory effects on skin by stabilizing lysosomes and reducing prostaglandin E2 synthesis.

Among the many cosmeceuticals, there is increasing interest in botanical antioxidants. Representatively, green tea polyphenols (GTPs) have gained attention because of their potent antioxidant activities. GTPs has been found to inhibit metalloproteinases and reduce the creation of cyclobutane pyrimidine dimers that are activated by UVR. Other botanic extracts have been described to contain potent antioxidant components include curcumin (turmeric), silymarin (milk thistle), apigenin, resveratrol (red grape), and genistein (soybean).

Cell regulators, such as vitamin A derivatives, polypeptides, hormones, GFs, and botanicals may act directly on the collagen metabolism and stimulate the production of collagen and elastic fibers. Polypeptides or oligopeptides are composed of amino acids and can imitate a peptide sequence of molecules such as collagen or elastin. Through topical application, polypeptides can stimulate collagen synthesis and activate dermal metabolism. A limited number of controlled studies demonstrate that topically applied GFs can stimulate collagen synthesis and epidermal thickening which is associated with clinical improvement in signs of photoaging. Because the use of GFs is still in an early stage, continued study to determine their efficacy is needed.

Alpha or beta hydroxy acids, whether natural or grafted with long-chain fatty acids, help smooth the skin. Fruit acids (malic, citric acids) have exfoliating actions on the epidermis and modulatory effects on fibroblasts in the dermis. Low concentrations of alpha hydroxy acids (AHAs) including glycolic and lactic acids are the most used in varying concentrations not normally exceeding 15%. They produce desquamation of the stratum corneum in the first weeks of use and improve the global appearance of the skin especially seborrheic skins and to a lesser extent the complexion, dyschromic lesions, and fine wrinkles. Beta hydroxy acids such as

salicylic acid produce superficial exfoliation on the skin surface and are mostly preferred for seborrheic skin with dilated pores.

More recently, products containing probiotics (mainly lysates of *Lactobacillus*, *Bifidobacterium*, *Vitreoscilla*, *Aquaphilus dolomiae*) have been used with the aim of strengthening the regulation of microbiota equilibrium, supporting skin barrier function, and preventing erythema.³⁴

5-fluorouracil and imiquimod

Topical 5-fluorouracil and imiquimod creams are effective for both AK and skin photoaging.^{35,36} Preparation-induced epidermal injury stimulates wound healing response in the skin with a subsequent dermal remodeling.

Systemic therapeutic agents with antiaging properties

Hormone replacement therapy

An acceleration of the skin aging process is observed in the postmenopausal period. Skin thickness decreases by 1.13% per year and the amount of collagen decreases by 2% per year after menopause. These detrimental changes can be mitigated with systemic or topical estrogen replacement therapy, which leads to increased epidermal hydration, increased skin elasticity, and improved vascularization. However, there are concerns regarding increased medical risks with its use. Thus, the US Preventive Services Task Force recommends against the use of hormone therapy in postmenopausal women for the primary prevention of chronic conditions.

Dietary supplements

Interventional studies indicate that it is in fact possible to delay skin aging and to improve skin conditions through administration of selected nutritional antioxidants.³⁷ A combination of antioxidant constituents is often used for antiaging including vitamins (A, E, C, flavonoid derivatives and carotene), trace elements (zinc,

selenium, copper, manganese, etc.), and plants (lycopene, soya derivatives, green tea, etc.) with limited evidence.³⁸

Hydrolyzed collagen supplementation demonstrated functional and beneficial effects on the skin, mainly by improving skin hydration, elasticity, and wrinkles.³⁹

Minimally invasive procedures

With the increased popularity of esthetic procedures to help make the skin appear more youthful, minimally invasive procedures including cryotherapy, chemical peels, botulinum toxin injections, soft-tissue augmentation with fillers, lasers, and other energy-based devices can be used to combat signs of aging.⁴⁰

Cryotherapy

Cryotherapy consists of applying liquid nitrogen to the lesion being treated. It is effective for solar lentigines and both seborrheic and actinic keratoses.

Chemical peels

Chemical peels are methods to cause a chemical ablation of defined skin layers to induce an even and tight skin as a result of the regeneration and repair mechanisms after the inflammation of the epidermis and dermis.⁴¹ The depth of peel required will depend on the location of the pathological process. Superficial peels can improve the appearance of the skin, medium-depth peels (to the upper reticular dermis) can treat superficial wrinkles and pigmentary changes, and deep peels (to the mid-reticular dermis) can treat deep wrinkles but are associated with significantly longer recovery times. Chemical peels have also been shown to be effective on AKs.⁴²

Neurotoxins and dermal fillers

Botulinum toxin is an injectable neuromodulator used to reduce hyperkinetic lines associated with muscles of facial expression.⁴³ The toxin functions by inhibiting acetylcholine

release at the neuromuscular junction, resulting in flaccid paralysis of targeted muscles. Furthermore, botulinum toxin can also stimulate the expression of type I collagen in human dermal fibroblasts, suggesting that botulinum toxin can stimulate remodeling of the ECM, which is essential for skin rejuvenation.⁴⁴

Dermal filler injections are used to improve coarse wrinkles and gradual loss of tissue volume.⁴⁵ Several studies demonstrated that fibroblasts in aged skin could be “rejuvenated” by enhancing structural support and mechanical force by injection of dermal filler, cross-linked HA into the skin. In addition, injection of dermal HA filler can stimulate fibroblasts to produce type I collagen associated with increase in mechanical forces. Enhanced mechanical support of the ECM also stimulates the proliferation of fibroblasts, endothelial cells, and keratinocytes, resulting in increased vasculature and epidermal thickness mediated by upregulation of TGF- β receptor and various GFs. In recent years, combination treatment with botulinum toxin and fillers was recommended to achieve superior clinical outcomes and greater patient satisfaction.

Additionally, microinjections of products containing one or multiple active ingredients including HA, vitamins, minerals, nutrients, hormones, GFs, amino acids, autologous cultured fibroblasts, homeopathic products such as platelet-rich plasma, and so on in the superficial dermis may stimulate the biosynthetic capacity of fibroblasts.^{46,47} However, more detailed studies are required to elucidate the exact mechanism of action and long-term efficiency of these procedures.

Lasers and other energy-based devices

The role of lasers in facial rejuvenation is well established and a range of lasers has been utilized in the treatment of aged skin.⁴⁸ Skin rejuvenation with laser technology is based on the concept that selective thermal damage to the dermis through laser energy can induce collagen

remodeling and tightening. Nonablative, ablative, and fractional laser systems are commonly used to improve rhytids, dyspigmentation, skin laxity, and other signs of photoaging.⁴⁹ In addition, radiofrequency (RF) devices and high-intensity focused ultrasound systems are used for skin tightening; pico- or nanosecond lasers and intense pulse light (IPL) systems are indicated for lentigines and as well as rejuvenation; vascular lasers and IPL systems are used for the treatment of photodamage-induced telangiectasias.⁵⁰

Future treatment options

The field of cosmetic dermatology continues to expand worldwide. Good understanding of the complexities of facial aging will pave the way for new treatment options. Current research may potentially clarify the genetic basis of skin youthfulness and may help to direct future therapies to target key biologic pathways of aging.^{51–53} It will be also possible to slow down the aging process through technological progress and innovation in the developing fields of tissue engineering and gene therapy.

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